NOM Outcomes - EDA

2026-01-25

The study dataset was subsetted to only individuals who received NOM at index. Of the 381 who received NOM, 41 (11%) received elective appendectomy and 89 (24%) received an urgent/emergent appendectomy. Limiting this to just one year follow-up from index, 39 (10%) received elective appendectomy and 80 (21%) received an urgent/emergent appendectomy.

```
## 
##            None        Elective Urgent/Emergent 
##             251              41              89
```

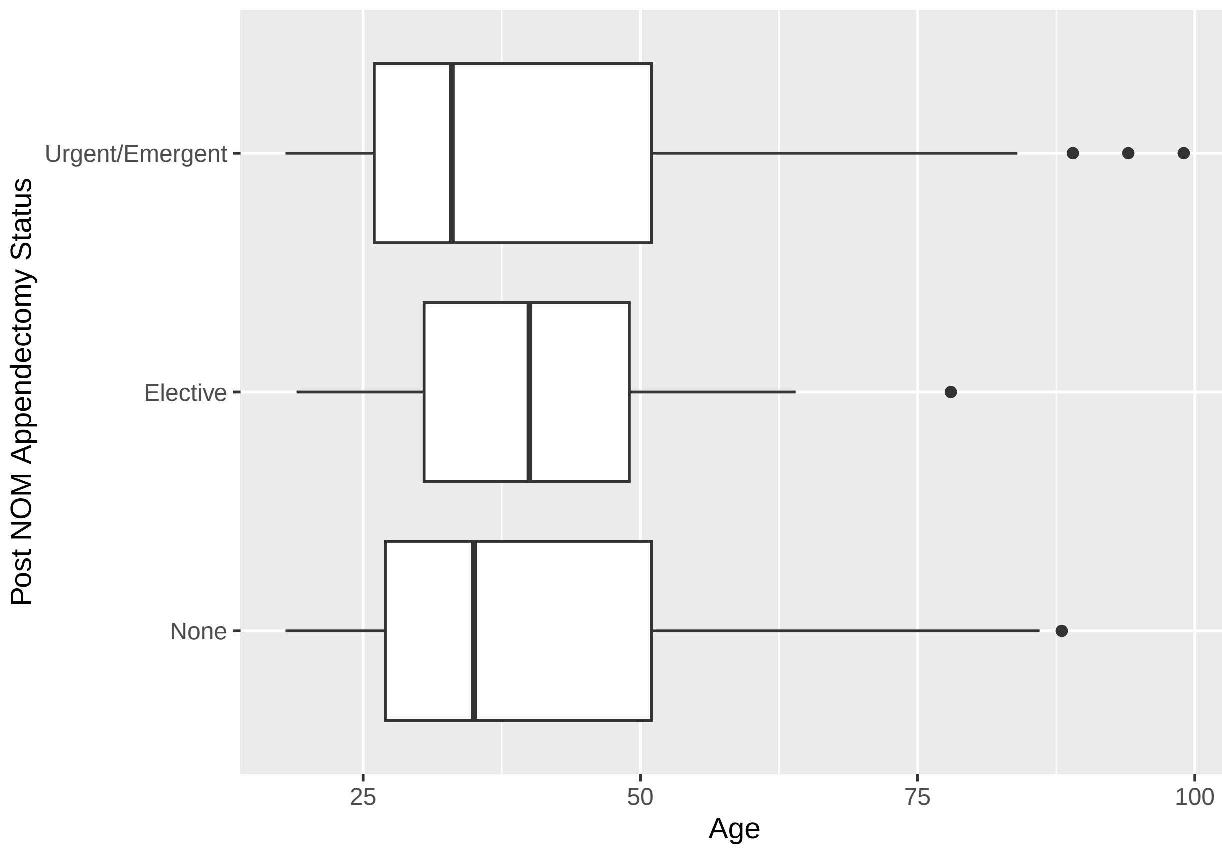
```
## 
##            None        Elective Urgent/Emergent 
##             262              39              80
```

RaceEthnicity has the most categories out of our potential covariates. When cross-tabulating these categories against post-NOM appendectomy status, there are many sparse or empty cells. If we were to use RaceEthnicity in a multivariate model, we would have to collapse some categories to prevent quasi-complete separation. Due to the sparseness, Fisher's exact test was used to test for an association. A p-value of 0.4608 indicates there is not a statistically significant relationship between Race/Ethnicity and post-NOM appendectomy status.

```
## 
## 
##    Cell Contents
## |-------------------------|
## |                       N |
## | Chi-square contribution |
## |           N / Row Total |
## |           N / Col Total |
## |         N / Table Total |
## |-------------------------|
## 
## 
## Total Observations in Table:  381 
## 
## 
##                                 | NOM_patients$status 
##     NOM_patients$RaceEthnicity |            None |        Elective | Urgent/Emergent |       Row Tota
## -------------------------------|-----------------|-----------------|-----------------|---------------
##                          Asian |              26 |               1 |               7 |              3
##                                |           0.579 |           1.932 |           0.112 |               
##                                |           0.765 |           0.029 |           0.206 |           0.08
##                                |           0.104 |           0.024 |           0.079 |               
##                                |           0.068 |           0.003 |           0.018 |               
## -------------------------------|-----------------|-----------------|-----------------|---------------
##         Black/African American |              40 |               6 |              13 |              5
##                                |           0.033 |           0.019 |           0.044 |               
##                                |           0.678 |           0.102 |           0.220 |           0.15
##                                |           0.159 |           0.146 |           0.146 |               
```

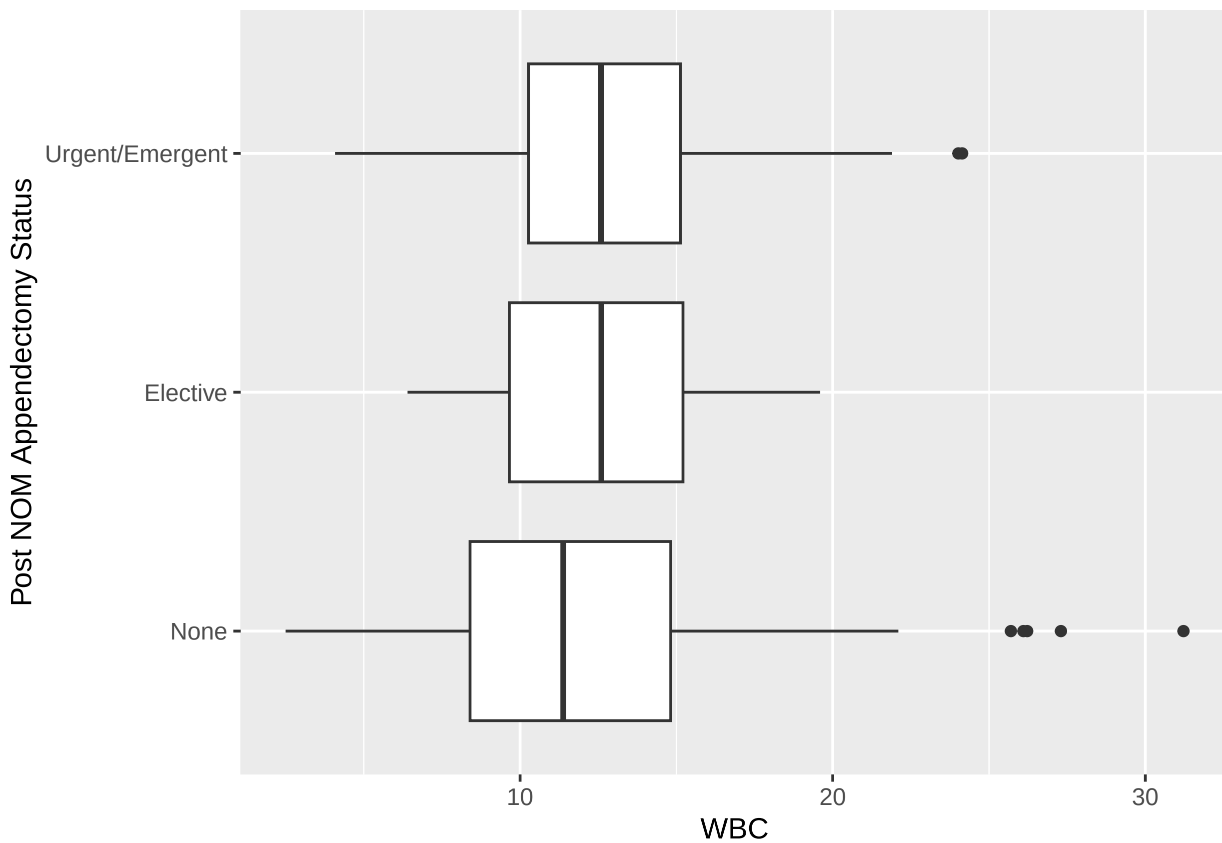
```
##                                 |            0.105 |            0.016 |            0.034 |
## --------------------------------|------------------|------------------|------------------|----------------
## Hispanic/Latino/Spanish Origin  |               52 |                8 |               27 |               8
##                                 |            0.493 |            0.198 |            2.194 |
##                                 |            0.598 |            0.092 |            0.310 |            0.22
##                                 |            0.207 |            0.195 |            0.303 |
##                                 |            0.136 |            0.021 |            0.071 |
## --------------------------------|------------------|------------------|------------------|----------------
##                     Multiracial |                8 |                0 |                4 |               1
##                                 |            0.001 |            1.291 |            0.511 |
##                                 |            0.667 |            0.000 |            0.333 |            0.03
##                                 |            0.032 |            0.000 |            0.045 |
##                                 |            0.021 |            0.000 |            0.010 |
## --------------------------------|------------------|------------------|------------------|----------------
##              Non-Hispanic White |              110 |               24 |               33 |              16
##                                 |            0.000 |            2.023 |            0.926 |
##                                 |            0.659 |            0.144 |            0.198 |            0.43
##                                 |            0.438 |            0.585 |            0.371 |
##                                 |            0.289 |            0.063 |            0.087 |
## --------------------------------|------------------|------------------|------------------|----------------
##             Other/Not Specified |               15 |                2 |                5 |               2
##                                 |            0.018 |            0.057 |            0.004 |
##                                 |            0.682 |            0.091 |            0.227 |            0.05
##                                 |            0.060 |            0.049 |            0.056 |
##                                 |            0.039 |            0.005 |            0.013 |
## --------------------------------|------------------|------------------|------------------|----------------
##                    Column Total |              251 |               41 |               89 |              38
##                                 |            0.659 |            0.108 |            0.234 |
## --------------------------------|------------------|------------------|------------------|----------------
##
##

##
##  Fisher's Exact Test for Count Data with simulated p-value (based on
##  2000 replicates)
##
## data:  cont_table
## p-value = 0.4843
## alternative hypothesis: two.sided
```

One-way ANOVA tests were used to test the association between post-NOM appendectomy status and continuous variables like age, WBC, and appendix diameter. All of these were found to not be significantly associated with post-NOM appendectomy status. Due to some sparse cells, Fisher's exact test was used to test the association between CCI and post-NOM appendectomy status and this test was also not statistically significant. Lastly, chi-squared tests were used to test the association between post-NOM appendectomy status and categorical variables like appendicolith presence and sex. Both these showed a not statistically significant relationship.




```
##              Df Sum Sq Mean Sq F value Pr(>F)
## status        2     34   17.02   0.055  0.946
## Residuals   378 116056  307.03
```

```
## Warning: Removed 1 row containing non-finite outside the scale range
## (`stat_boxplot()`).
```



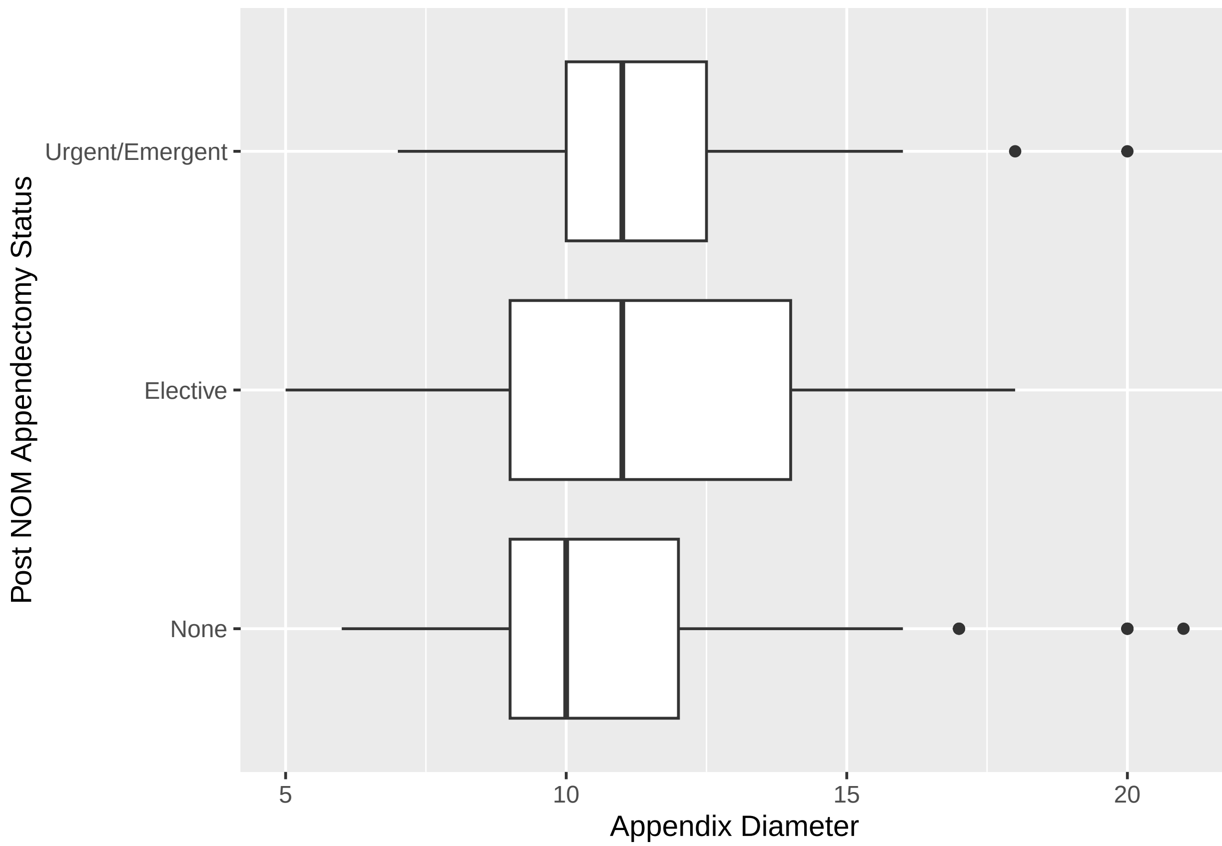

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## status        2     41   20.32   0.971   0.38
## Residuals   377   7891   20.93
## 1 observation deleted due to missingness
```

```
##
##
##    Cell Contents
## |-------------------------|
## |                       N |
## | Chi-square contribution |
## |           N / Row Total |
## |           N / Col Total |
## |         N / Table Total |
## |-------------------------|
##
##
## Total Observations in Table:  376
##
##
##                         | NOM_patients$status
## NOM_patients$ccioutcomes |             None |         Elective | Urgent/Emergent |        Row Total |
## -------------------------|------------------|------------------|-----------------|------------------|
##                     >= 8 |                3 |                0 |               0 |                3 |
##                         |            0.527 |            0.319 |           0.702 |                  |
##                         |            1.000 |            0.000 |           0.000 |            0.008 |
##                         |            0.012 |            0.000 |           0.000 |                  |
```

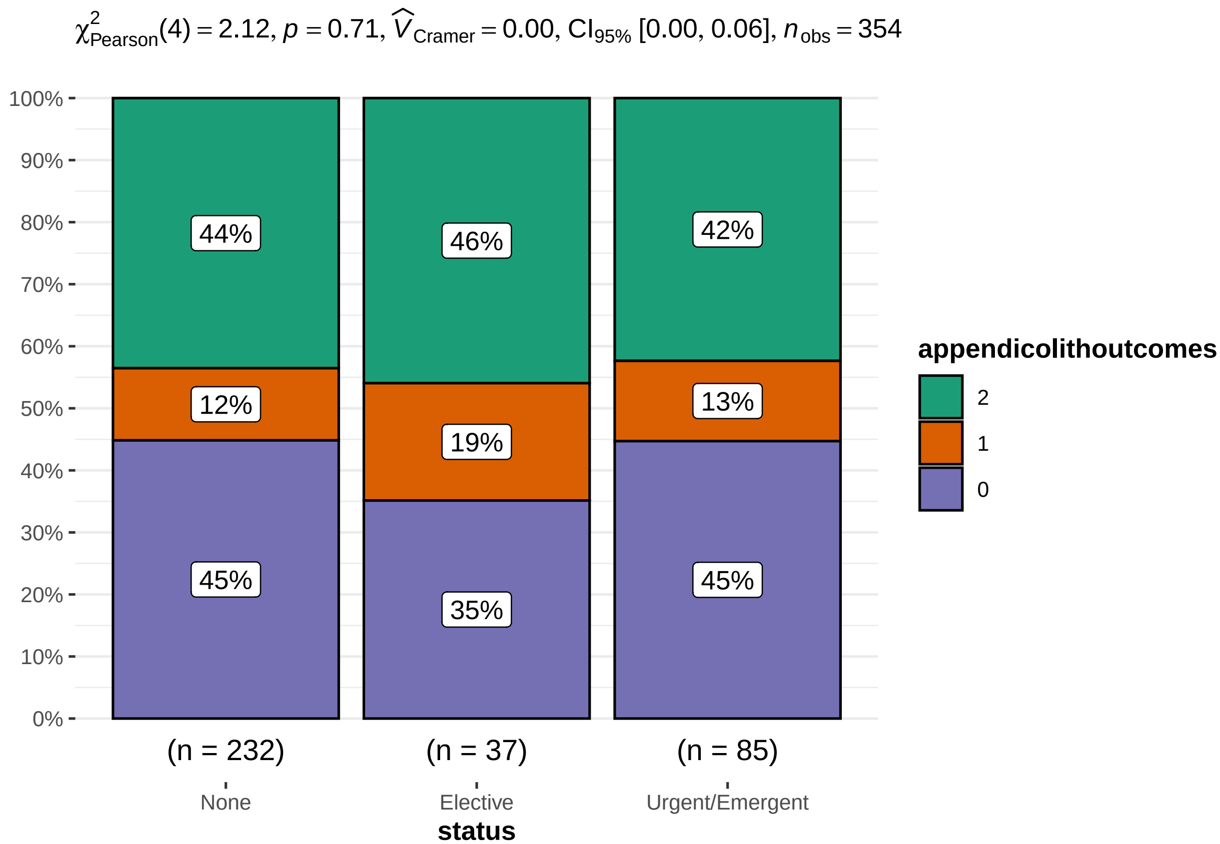
```
##                          |        0.008 |        0.000 |        0.000 |              |
## -------------------------|--------------|--------------|--------------|--------------|
##                        0 |          173 |           28 |           63 |          264 |
##                          |        0.007 |        0.000 |        0.024 |              |
##                          |        0.655 |        0.106 |        0.239 |        0.702 |
##                          |        0.698 |        0.700 |        0.716 |              |
##                          |        0.460 |        0.074 |        0.168 |              |
## -------------------------|--------------|--------------|--------------|--------------|
##                        1 |           29 |            6 |            8 |           43 |
##                          |        0.014 |        0.444 |        0.423 |              |
##                          |        0.674 |        0.140 |        0.186 |        0.114 |
##                          |        0.117 |        0.150 |        0.091 |              |
##                          |        0.077 |        0.016 |        0.021 |              |
## -------------------------|--------------|--------------|--------------|--------------|
##                        2 |           15 |            4 |            6 |           25 |
##                          |        0.135 |        0.676 |        0.004 |              |
##                          |        0.600 |        0.160 |        0.240 |        0.066 |
##                          |        0.060 |        0.100 |        0.068 |              |
##                          |        0.040 |        0.011 |        0.016 |              |
## -------------------------|--------------|--------------|--------------|--------------|
##                        3 |           10 |            0 |            4 |           14 |
##                          |        0.064 |        1.489 |        0.160 |              |
##                          |        0.714 |        0.000 |        0.286 |        0.037 |
##                          |        0.040 |        0.000 |        0.045 |              |
##                          |        0.027 |        0.000 |        0.011 |              |
## -------------------------|--------------|--------------|--------------|--------------|
##                        4 |           13 |            1 |            2 |           16 |
##                          |        0.567 |        0.290 |        0.813 |              |
##                          |        0.812 |        0.062 |        0.125 |        0.043 |
##                          |        0.052 |        0.025 |        0.023 |              |
##                          |        0.035 |        0.003 |        0.005 |              |
## -------------------------|--------------|--------------|--------------|--------------|
##                        5 |            2 |            1 |            2 |            5 |
##                          |        0.511 |        0.412 |        0.588 |              |
##                          |        0.400 |        0.200 |        0.400 |        0.013 |
##                          |        0.008 |        0.025 |        0.023 |              |
##                          |        0.005 |        0.003 |        0.005 |              |
## -------------------------|--------------|--------------|--------------|--------------|
##                        6 |            1 |            0 |            2 |            3 |
##                          |        0.484 |        0.319 |        2.399 |              |
##                          |        0.333 |        0.000 |        0.667 |        0.008 |
##                          |        0.004 |        0.000 |        0.023 |              |
##                          |        0.003 |        0.000 |        0.005 |              |
## -------------------------|--------------|--------------|--------------|--------------|
##                        7 |            2 |            0 |            1 |            3 |
##                          |        0.000 |        0.319 |        0.126 |              |
##                          |        0.667 |        0.000 |        0.333 |        0.008 |
##                          |        0.008 |        0.000 |        0.011 |              |
##                          |        0.005 |        0.000 |        0.003 |              |
## -------------------------|--------------|--------------|--------------|--------------|
##             Column Total |          248 |           40 |           88 |          376 |
##                          |        0.660 |        0.106 |        0.234 |              |
## -------------------------|--------------|--------------|--------------|--------------|
##
```

```
##

##
##  Fisher's Exact Test for Count Data
##
## data:  contingency_table
## p-value = 0.7517
## alternative hypothesis: two.sided
```

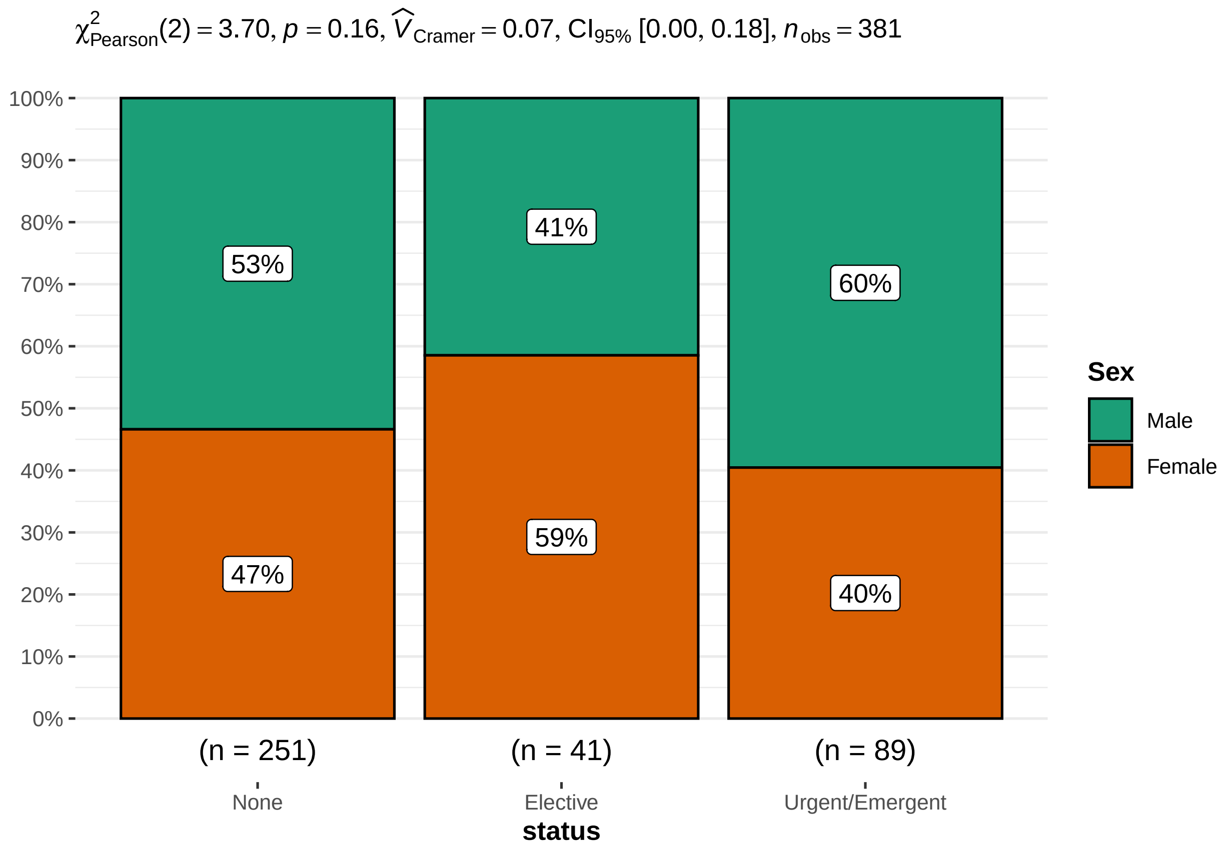
```
## Warning: Removed 93 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```




```
##              Df Sum Sq Mean Sq F value Pr(>F)
## status        2   15.5   7.728   1.035  0.357
## Residuals   285 2128.4   7.468
## 93 observations deleted due to missingness
```



$\chi^2_{\text{Pearson}}(4) = 2.12$, $p = 0.71$, $\widehat{V}_{\text{Cramer}} = 0.00$, $\text{CI}_{95\%}$ [0.00, 0.06], $n_{\text{obs}} = 354$
100%
90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
44%
12%
45%
46%
19%
35%
42%
13%
45%
(n = 232)
(n = 37)
(n = 85)
None
Elective
Urgent/Emergent
status
appendicolithoutcomes
2
1
0



$\chi^2_{\text{Pearson}}(2) = 3.70$, $p = 0.16$, $\widehat{V}_{\text{Cramer}} = 0.07$, $\text{CI}_{95\%}$ [0.00, 0.18], $n_{\text{obs}} = 381$
100%
90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
53%
47%
41%
59%
60%
40%
(n = 251)
(n = 41)
(n = 89)
None
Elective
Urgent/Emergent
status
Sex
Male
Female